

hPTM signatures association with gene age and expression dynamics

Here, I load the supplementary table from Jeromine's hPTM signatures and examine the association between the signatures and evolutionary (as captured by gene age) and expression dynamics (as captured by tau)

```
library(tidyverse)
library(ggrrastr)
library(readxl)
library(scales)

readxl::excel_sheets("data/dataset_S3_master_tables.xlsx")

## [1] "Ectocarpus"          "Desmarestia-dudresnayi" "Desmarestia-herbacea"
## [4] "Undaria"            "Scytosiphon"          "Schizocladia"

Ec_ps <- readxl::read_xlsx("data/dataset_S3_master_tables.xlsx", sheet = "Ectocarpus") |>
  dplyr::select(gene_id, dplyr::contains("TPM"), dplyr::contains("Signature"), rank)
Ddud_ps <- readxl::read_xlsx("data/dataset_S3_master_tables.xlsx", sheet = "Desmarestia-dudresnayi") |>
  dplyr::select(gene_id, dplyr::contains("TPM"), dplyr::contains("Signature"), rank)
Dher_ps <- readxl::read_xlsx("data/dataset_S3_master_tables.xlsx", sheet = "Desmarestia-herbacea") |>
  dplyr::select(gene_id, dplyr::contains("TPM"), dplyr::contains("Signature"), rank)
Und_ps <- readxl::read_xlsx("data/dataset_S3_master_tables.xlsx", sheet = "Undaria") |>
  dplyr::select(gene_id, dplyr::contains("TPM"), dplyr::contains("Signature"), rank)
Spro_ps <- readxl::read_xlsx("data/dataset_S3_master_tables.xlsx", sheet = "Scytosiphon") |>
  dplyr::select(gene_id, dplyr::contains("TPM"), dplyr::contains("Signature"), rank)
Si_ps <- readxl::read_xlsx("data/dataset_S3_master_tables.xlsx", sheet = "Schizocladia") |>
  dplyr::select(gene_id, dplyr::contains("TPM"), dplyr::contains("Signature"), rank)
```

Note: Scytosiphon is misspelt in the table

```
preprocess_ps_sig <- function(Species_ps){
  Species_ps.filt <- Species_ps |>
    tidyr::drop_na() |>
    dplyr::mutate(rank = as.numeric(rank)) |>
    tidyr::drop_na(rank)
  if(any(colnames(Species_ps.filt) %in% c("Signature.Female", "Signature.Male"))){
    Species_ps.filt2 <-
      Species_ps.filt |>
      dplyr::mutate(Signature.Female = factor(
        Signature.Female, levels = paste0('S',1:16))) |>
      dplyr::mutate(Signature.Male = factor(
        Signature.Male, levels = paste0('S',1:16)))
  }else{
    Species_ps.filt2 <-
      Species_ps.filt |>
      dplyr::mutate(Signature = factor(
        Signature, levels = paste0('S',1:16)))
  }
  return(Species_ps.filt2)
```

```

}

color_table <-
  tibble::tibble(
    Signature = paste0('S',1:16),
    Colour = c("#a31d21", "#e43a30", "#f36f57", "#f8a68d", "#fddfd1", "#1b733b",
               "#2daf66", "#6ec283", "#aed9b1", "#9392c8", "#bfbfd", "#e9eef8",
               "#2a74bb", "#61abd7", "#afd1e5", "#999999")
  )

plot_style <- function(p_in){
  p_out <-
    p_in +
    ggtrastr::rasterise(ggplot2::geom_jitter(width = 0.3, alpha = 0.1),
                        dpi = 200) +
    ggplot2::geom_boxplot(width = 0.5, size = 0.8, outliers = FALSE, colour = "black",
                          linewidth = 0.4) +
    ggplot2::theme_classic() +
    ggplot2::scale_colour_manual(values = rev(color_table$Colour)) +
    ggplot2::theme(legend.position = "none") +
    ggplot2::coord_flip() +
    # ggplot2::scale_x_discrete(limits=rev) +
    ggplot2::scale_y_continuous(breaks = scales::breaks_pretty())
  return(p_out)
}

```

plotting signatures

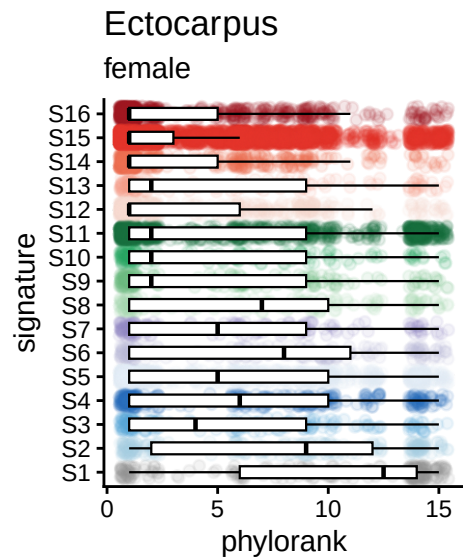
hPTM signatures and mean expression or gene age.

Ectocarpus

```

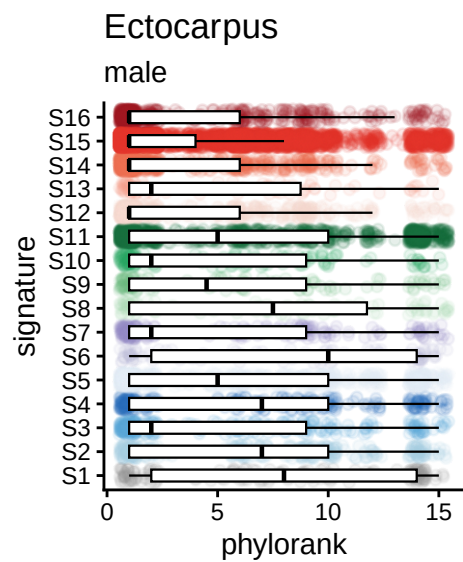
Ec_ps.filt <-
  Ec_ps |>
  preprocess_ps_sig()
Ec_ps.filt |>
  dplyr::filter(TPM.Female > 1 | TPM.Male > 1) |>
  ggplot2::ggplot(aes(x = Signature.Female, colour = Signature.Female, y = rank)) |>
  plot_style() +
  ggplot2::labs(x = "signature", y = "phylorank",
                title = "Ectocarpus", subtitle = "female")

```



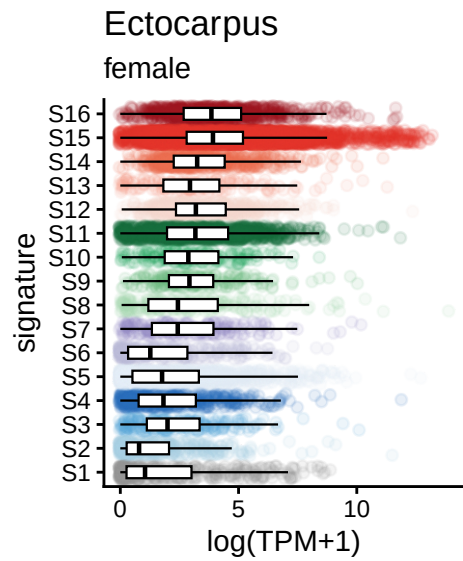
```
# ggplot2::ggsave("figure/signature_phylorank/Ec_rank_rev2.pdf", width = 3, height = 3)
```

```
Ec_ps.filt |>
  dplyr::filter(TPM.Female > 1 | TPM.Male > 1) |>
  ggplot2::ggplot(aes(x = Signature.Male, colour = Signature.Male, y = rank)) |>
  plot_style() +
  ggplot2::labs(x = "signature", y = "phylorank",
                title = "Ectocarpus", subtitle = "male")
```

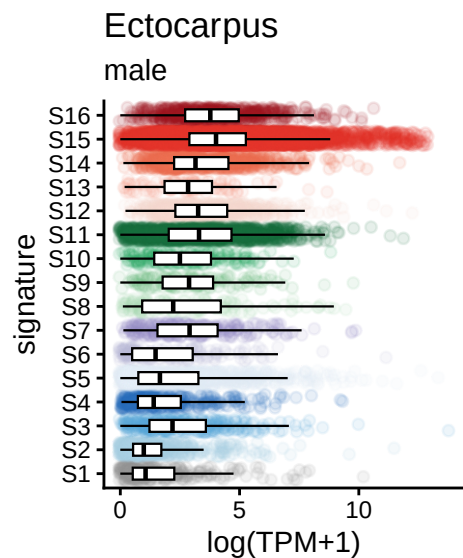


```
# expression
```

```
Ec_ps.filt |>
  ggplot2::ggplot(aes(x = Signature.Female, colour = Signature.Female, y = log2_TPM_plus_1.Female)) |>
  plot_style() +
  ggplot2::labs(x = "signature", y = "log(TPM+1)",
                title = "Ectocarpus", subtitle = "female")
```



```
Ec_ps.filt |>
  ggplot2::ggplot(aes(x = Signature.Male, colour = Signature.Male, y = log2_TPM_plus_1.Male)) |>
  plot_style() +
  ggplot2::labs(x = "signature", y = "log(TPM+1)",
                title = "Ectocarpus", subtitle = "male")
```



```
kruskal.test(log2_TPM_plus_1.Female ~ Signature.Female, data = Ec_ps.filt)
```

```
##
##  Kruskal-Wallis rank sum test
##
## data:  log2_TPM_plus_1.Female by Signature.Female
## Kruskal-Wallis chi-squared = 3110.9, df = 15, p-value < 2.2e-16
```

```
kruskal.test(log2_TPM_plus_1.Male ~ Signature.Male, data = Ec_ps.filt)
```

```
##
##  Kruskal-Wallis rank sum test
##
```

```
## data: log2_TPM_plus_1.Male by Signature.Male
## Kruskal-Wallis chi-squared = 3452.5, df = 15, p-value < 2.2e-16
```

```
kruskal.test(rank ~ Signature.Female, data = Ec_ps.filt |>
  dplyr::filter(TPM.Female > 1 | TPM.Male > 1))
```

```
##
## Kruskal-Wallis rank sum test
##
## data: rank by Signature.Female
## Kruskal-Wallis chi-squared = 1595.5, df = 15, p-value < 2.2e-16
```

```
kruskal.test(rank ~ Signature.Male, data = Ec_ps.filt |>
  dplyr::filter(TPM.Female > 1 | TPM.Male > 1))
```

```
##
## Kruskal-Wallis rank sum test
##
## data: rank by Signature.Male
## Kruskal-Wallis chi-squared = 1333.3, df = 15, p-value < 2.2e-16
```

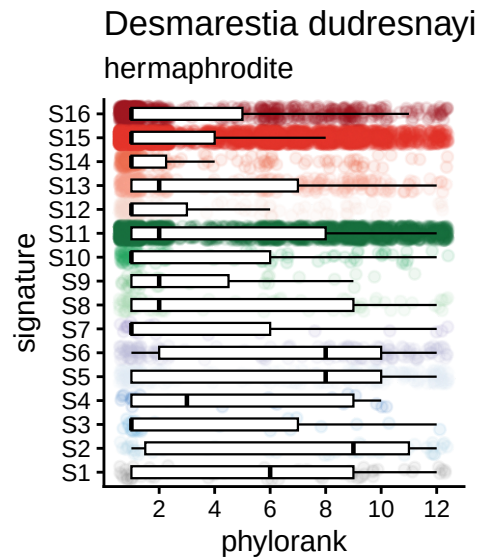
In terms of information, the grouping reduces entropy w.r.t. gene age ranks.

```
# for male signatures
Marginal entropy (no groups): 2.683 bits
Conditional entropy (with groups): 2.547 bits
Mutual information (bits gained by knowing group): 0.135 bits
```

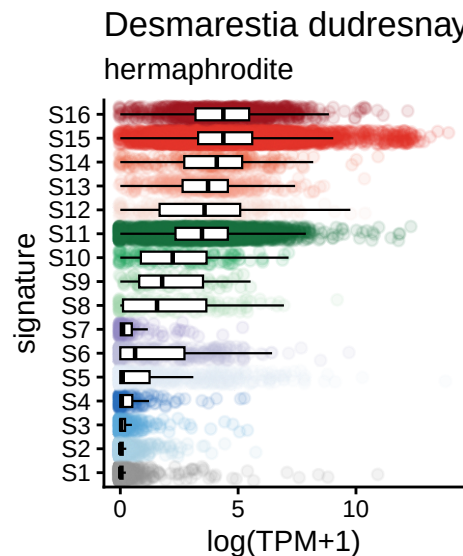
```
# for female signatures
Marginal entropy (no groups): 2.683 bits
Conditional entropy (with groups): 2.529 bits
Mutual information (bits gained by knowing group): 0.154 bits
```

Desmarestia-dudresnayi

```
Ddud_ps.filt <-
  Ddud_ps |>
  preprocess_ps_sig()
Ddud_ps.filt |>
  dplyr::filter(TPM > 1) |>
  ggplot2::ggplot(aes(x = Signature, colour = Signature, y = rank)) |>
  plot_style() +
  ggplot2::labs(x = "signature", y = "phylorank",
    title = "Desmarestia dudresnayi", subtitle = "hermaphrodite")
```



```
# ggplot2::ggsave("figure/signature_phylorank/Ddud_rank_rev2.pdf", width = 3, height = 3)
# expression
Ddud_ps.filt |>
  ggplot2::ggplot(aes(x = Signature, colour = Signature, y = log2_TPM_plus_1)) |>
  plot_style() +
  ggplot2::labs(x = "signature", y = "log(TPM+1)",
                title = "Desmarestia dudresnayi", subtitle = "hermaphrodite")
```



```
kruskal.test(log2_TPM_plus_1 ~ Signature, data = Ddud_ps.filt)

##
## Kruskal-Wallis rank sum test
##
## data: log2_TPM_plus_1 by Signature
## Kruskal-Wallis chi-squared = 9381.9, df = 15, p-value < 2.2e-16

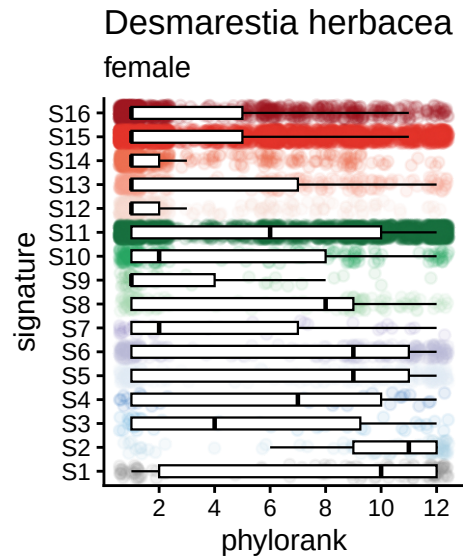
kruskal.test(rank ~ Signature, data = Ddud_ps.filt |>
  dplyr::filter(TPM > 1))
```

```
##
## Kruskal-Wallis rank sum test
##
## data: rank by Signature
## Kruskal-Wallis chi-squared = 897.39, df = 15, p-value < 2.2e-16
```

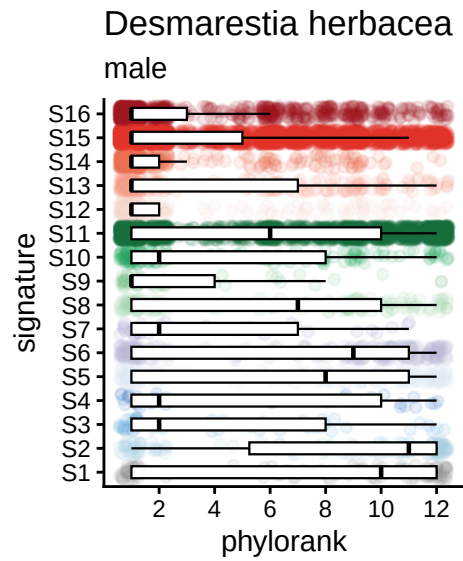
Marginal **entropy** (no groups): 2.732 bits
Conditional **entropy** (with groups): 2.587 bits
Mutual **information** (bits gained by knowing group): 0.145 bits

Desmarestia-herbacea

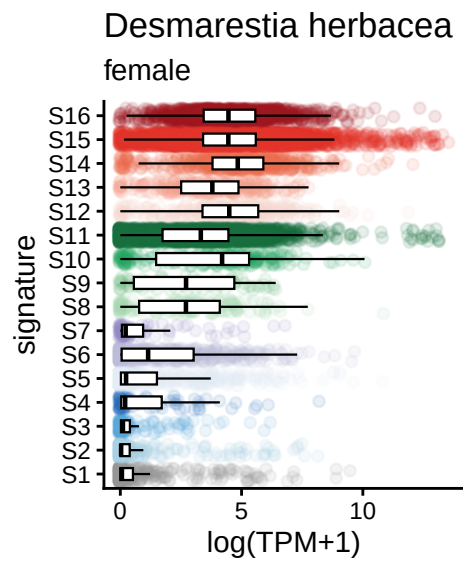
```
Dher_ps.filt <-
  Dher_ps |>
  preprocess_ps_sig()
Dher_ps.filt |>
  dplyr::filter(TPM.Female > 1 | TPM.Male > 1) |>
  ggplot2::ggplot(aes(x = Signature.Female, colour = Signature.Female, y = rank)) |>
  plot_style() +
  ggplot2::labs(x = "signature", y = "phylorank",
               title = "Desmarestia herbacea", subtitle = "female")
```



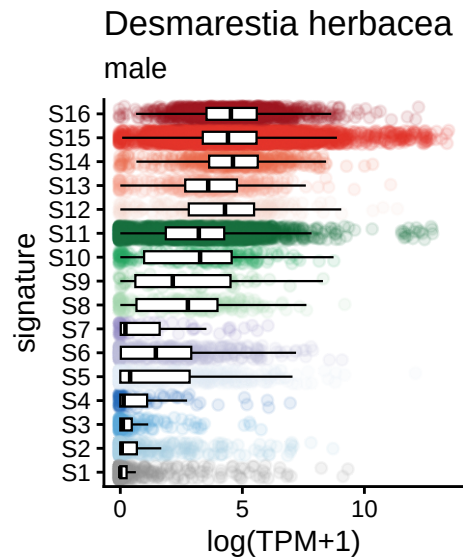
```
# ggplot2::ggsave("figure/signature_phylorank/Dher_rank_rev2.pdf", width = 3, height = 3)
Dher_ps.filt |>
  dplyr::filter(TPM.Female > 1 | TPM.Male > 1) |>
  ggplot2::ggplot(aes(x = Signature.Male, colour = Signature.Male, y = rank)) |>
  plot_style() +
  ggplot2::labs(x = "signature", y = "phylorank",
               title = "Desmarestia herbacea", subtitle = "male")
```



```
# expression
Dher_ps.filt |>
  ggplot2::ggplot(aes(x = Signature.Female, colour = Signature.Female, y = log2_TPM_plus_1.Female)) |>
  plot_style() +
  ggplot2::labs(x = "signature", y = "log(TPM+1)",
    title = "Desmarestia herbacea", subtitle = "female")
```



```
Dher_ps.filt |>
  ggplot2::ggplot(aes(x = Signature.Male, colour = Signature.Male, y = log2_TPM_plus_1.Male)) |>
  plot_style() +
  ggplot2::labs(x = "signature", y = "log(TPM+1)",
    title = "Desmarestia herbacea", subtitle = "male")
```



```
kruskal.test(log2_TPM_plus_1.Female ~ Signature.Female, data = Dher_ps.filt)
```

```
##
## Kruskal-Wallis rank sum test
##
## data: log2_TPM_plus_1.Female by Signature.Female
## Kruskal-Wallis chi-squared = 4836.9, df = 15, p-value < 2.2e-16
```

```
kruskal.test(log2_TPM_plus_1.Male ~ Signature.Male, data = Dher_ps.filt)
```

```
##
## Kruskal-Wallis rank sum test
##
## data: log2_TPM_plus_1.Male by Signature.Male
## Kruskal-Wallis chi-squared = 5344.4, df = 15, p-value < 2.2e-16
```

```
kruskal.test(rank ~ Signature.Female, data = Dher_ps.filt |>
  dplyr::filter(TPM.Female > 1 | TPM.Male > 1))
```

```
##
## Kruskal-Wallis rank sum test
##
## data: rank by Signature.Female
## Kruskal-Wallis chi-squared = 1946.5, df = 15, p-value < 2.2e-16
```

```
kruskal.test(rank ~ Signature.Male, data = Dher_ps.filt |>
  dplyr::filter(TPM.Female > 1 | TPM.Male > 1))
```

```
##
## Kruskal-Wallis rank sum test
##
## data: rank by Signature.Male
## Kruskal-Wallis chi-squared = 1928.7, df = 15, p-value < 2.2e-16
```

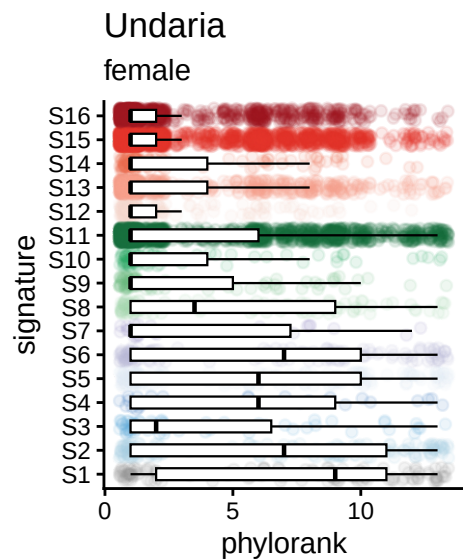
```
# for male
```

```
Marginal entropy (no groups): 2.632 bits
Conditional entropy (with groups): 2.496 bits
Mutual information (bits gained by knowing group): 0.136 bits
```

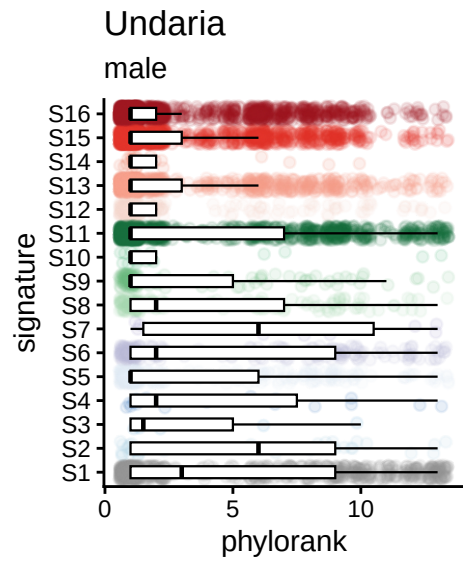
```
# for female
Marginal entropy (no groups): 2.632 bits
Conditional entropy (with groups): 2.493 bits
Mutual information (bits gained by knowing group): 0.139 bits
```

Undaria

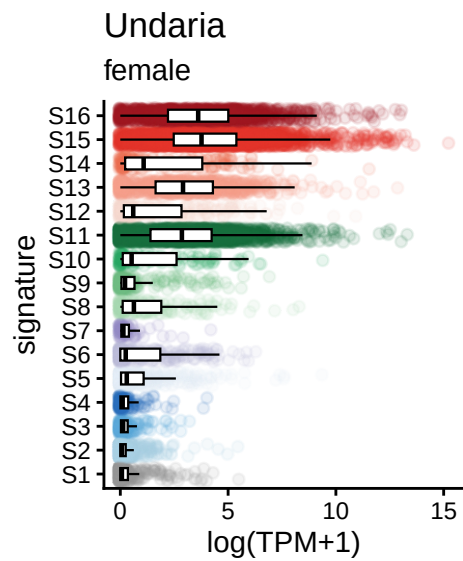
```
Und_ps.filt <-
  Und_ps |>
  preprocess_ps_sig()
Und_ps.filt |>
  dplyr::filter(TPM.Female > 1 | TPM.Male > 1) |>
  ggplot2::ggplot(aes(x = Signature.Female, colour = Signature.Female, y = rank)) |>
  plot_style() +
  ggplot2::labs(x = "signature", y = "phylorank",
               title = "Undaria", subtitle = "female")
```



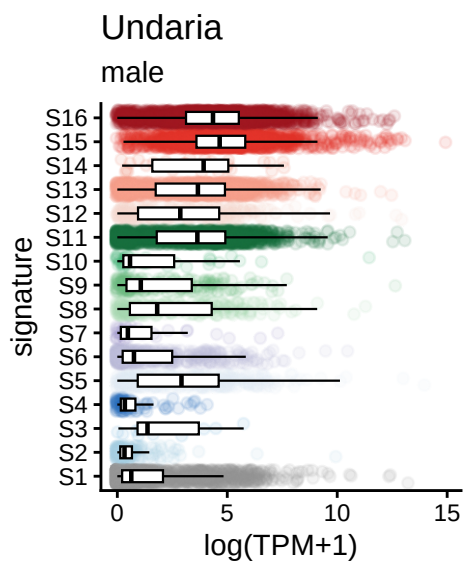
```
# ggplot2::ggsave("figure/signature_phylorank/Und_rank_rev2.pdf", width = 3, height = 3)
Und_ps.filt |>
  dplyr::filter(TPM.Female > 1 | TPM.Male > 1) |>
  ggplot2::ggplot(aes(x = Signature.Male, colour = Signature.Male, y = rank)) |>
  plot_style() +
  ggplot2::labs(x = "signature", y = "phylorank",
               title = "Undaria", subtitle = "male")
```



```
# expression
Und_ps.filt |>
  ggplot2::ggplot(aes(x = Signature.Female, colour = Signature.Female, y = log2_TPM_plus_1.Female)) |>
  plot_style() +
  ggplot2::labs(x = "signature", y = "log(TPM+1)",
    title = "Undaria", subtitle = "female")
```



```
Und_ps.filt |>
  ggplot2::ggplot(aes(x = Signature.Male, colour = Signature.Male, y = log2_TPM_plus_1.Male)) |>
  plot_style() +
  ggplot2::labs(x = "signature", y = "log(TPM+1)",
    title = "Undaria", subtitle = "male")
```



```
kruskal.test(log2_TPM_plus_1.Female ~ Signature.Female, data = Und_ps.filt)
```

```
##
## Kruskal-Wallis rank sum test
##
## data: log2_TPM_plus_1.Female by Signature.Female
## Kruskal-Wallis chi-squared = 4825.4, df = 15, p-value < 2.2e-16
```

```
kruskal.test(log2_TPM_plus_1.Male ~ Signature.Male, data = Und_ps.filt)
```

```
##
## Kruskal-Wallis rank sum test
##
## data: log2_TPM_plus_1.Male by Signature.Male
## Kruskal-Wallis chi-squared = 3356.1, df = 15, p-value < 2.2e-16
```

```
kruskal.test(rank ~ Signature.Female, data = Und_ps.filt |>
  dplyr::filter(TPM.Female > 1 | TPM.Male > 1))
```

```
##
## Kruskal-Wallis rank sum test
##
## data: rank by Signature.Female
## Kruskal-Wallis chi-squared = 604.32, df = 15, p-value < 2.2e-16
```

```
kruskal.test(rank ~ Signature.Male, data = Und_ps.filt |>
  dplyr::filter(TPM.Female > 1 | TPM.Male > 1))
```

```
##
## Kruskal-Wallis rank sum test
##
## data: rank by Signature.Male
## Kruskal-Wallis chi-squared = 479.27, df = 15, p-value < 2.2e-16
```

```
# male
```

```
Marginal entropy (no groups): 2.535 bits
```

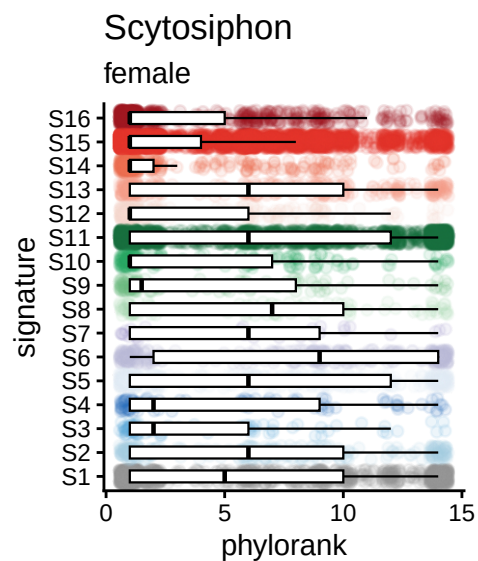
```
Conditional entropy (with groups): 2.432 bits
```

```
Mutual information (bits gained by knowing group): 0.103 bits
```

```
# female
Marginal entropy (no groups): 2.535 bits
Conditional entropy (with groups): 2.427 bits
Mutual information (bits gained by knowing group): 0.108 bits
```

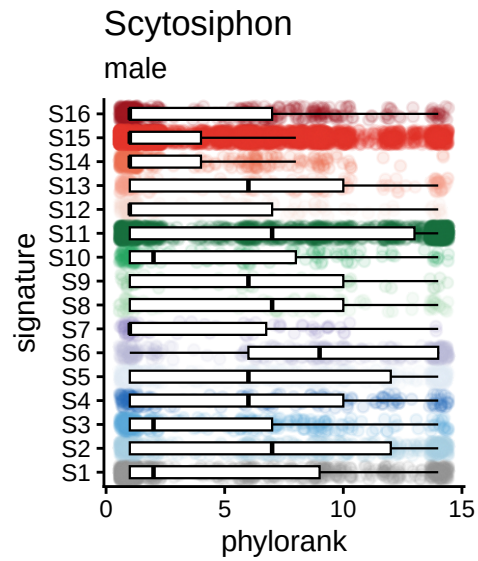
Scytosiphon

```
Spro_ps.filt <-
  Spro_ps |>
  preprocess_ps_sig()
Spro_ps.filt |>
  dplyr::filter(TPM.Female > 1 | TPM.Male > 1) |>
  ggplot2::ggplot(aes(x = Signature.Female, colour = Signature.Female, y = rank)) |>
  plot_style() +
  ggplot2::labs(x = "signature", y = "phylorank",
               title = "Scytosiphon", subtitle = "female")
```

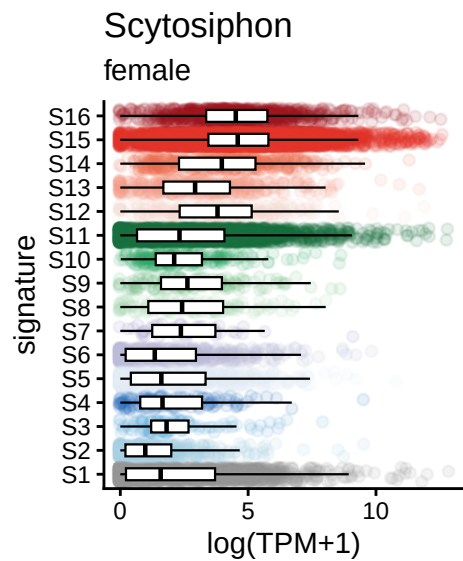


```
# ggplot2::ggsave("figure/signature_phylorank/Spro_rank_rev2.pdf", width = 3, height = 3)
```

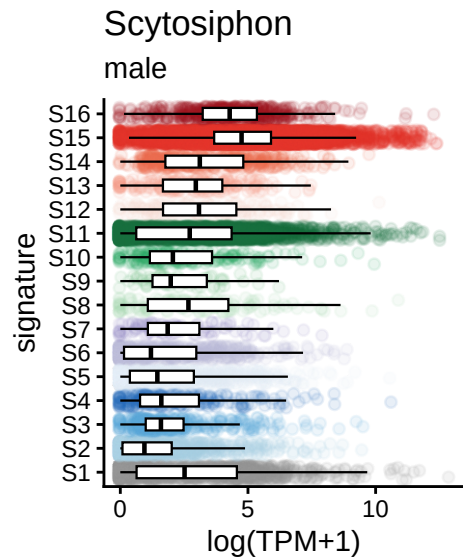
```
Spro_ps.filt |>
  dplyr::filter(TPM.Female > 1 | TPM.Male > 1) |>
  ggplot2::ggplot(aes(x = Signature.Male, colour = Signature.Male, y = rank)) |>
  plot_style() +
  ggplot2::labs(x = "signature", y = "phylorank",
               title = "Scytosiphon", subtitle = "male")
```



```
# expression
Spro_ps.filt |>
  ggplot2::ggplot(aes(x = Signature.Female, colour = Signature.Female, y = log2_TPM_plus_1.Female)) |>
  plot_style() +
  ggplot2::labs(x = "signature", y = "log(TPM+1)",
    title = "Scytosiphon", subtitle = "female")
```



```
Spro_ps.filt |>
  ggplot2::ggplot(aes(x = Signature.Male, colour = Signature.Male, y = log2_TPM_plus_1.Male)) |>
  plot_style() +
  ggplot2::labs(x = "signature", y = "log(TPM+1)",
    title = "Scytosiphon", subtitle = "male")
```



```
kruskal.test(log2_TPM_plus_1.Female ~ Signature.Female, data = Spro_ps.filt)
```

```
##
## Kruskal-Wallis rank sum test
##
## data: log2_TPM_plus_1.Female by Signature.Female
## Kruskal-Wallis chi-squared = 5210.4, df = 15, p-value < 2.2e-16
```

```
kruskal.test(log2_TPM_plus_1.Male ~ Signature.Male, data = Spro_ps.filt)
```

```
##
## Kruskal-Wallis rank sum test
##
## data: log2_TPM_plus_1.Male by Signature.Male
## Kruskal-Wallis chi-squared = 5908.6, df = 15, p-value < 2.2e-16
```

```
kruskal.test(rank ~ Signature.Female, data = Spro_ps.filt |>
  dplyr::filter(TPM.Female > 1 | TPM.Male > 1))
```

```
##
## Kruskal-Wallis rank sum test
##
## data: rank by Signature.Female
## Kruskal-Wallis chi-squared = 2098.9, df = 15, p-value < 2.2e-16
```

```
kruskal.test(rank ~ Signature.Male, data = Spro_ps.filt |>
  dplyr::filter(TPM.Female > 1 | TPM.Male > 1))
```

```
##
## Kruskal-Wallis rank sum test
##
## data: rank by Signature.Male
## Kruskal-Wallis chi-squared = 2275.1, df = 15, p-value < 2.2e-16
```

```
# male
```

```
Marginal entropy (no groups): 2.722 bits
```

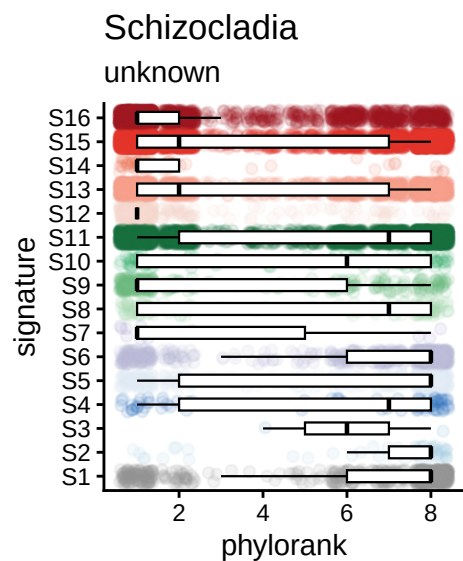
```
Conditional entropy (with groups): 2.517 bits
```

```
Mutual information (bits gained by knowing group): 0.205 bits
```

```
# female
Marginal entropy (no groups): 2.722 bits
Conditional entropy (with groups): 2.524 bits
Mutual information (bits gained by knowing group): 0.198 bits
```

Schizocladia

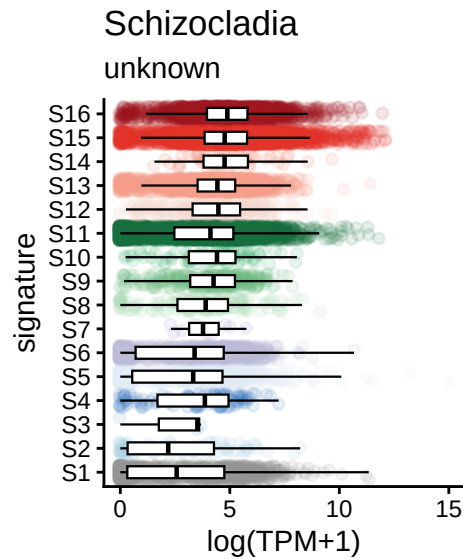
```
Si_ps.filt <-
  Si_ps |>
  preprocess_ps_sig()
Si_ps.filt |>
  dplyr::filter(TPM > 1) |>
  ggplot2::ggplot(aes(x = Signature, colour = Signature, y = rank)) |>
  plot_style() +
  ggplot2::labs(x = "signature", y = "phylorank",
                title = "Schizocladia", subtitle = "unknown")
```



```
# ggplot2::ggsave("figure/signature_phylorank/Si_rank_rev2.pdf", width = 3, height = 3)
```

```
# expression
```

```
Si_ps.filt |>
  ggplot2::ggplot(aes(x = Signature, colour = Signature, y = log2_TPM_plus_1)) |>
  plot_style() +
  ggplot2::labs(x = "signature", y = "log(TPM+1)",
                title = "Schizocladia", subtitle = "unknown")
```



```
kruskal.test(log2_TPM_plus_1 ~ Signature, data = Si_ps.filt)
```

```
##
## Kruskal-Wallis rank sum test
##
## data: log2_TPM_plus_1 by Signature
## Kruskal-Wallis chi-squared = 2071.8, df = 15, p-value < 2.2e-16
```

```
kruskal.test(rank ~ Signature, data = Si_ps.filt |>
  dplyr::filter(TPM > 1))
```

```
##
## Kruskal-Wallis rank sum test
##
## data: rank by Signature
## Kruskal-Wallis chi-squared = 3462.5, df = 15, p-value < 2.2e-16
```

```
Marginal entropy (no groups): 2.185 bits
Conditional entropy (with groups): 2.001 bits
Mutual information (bits gained by knowing group): 0.184 bits
```

Across all species, the MI is around 4 to 8%.

hPTM signatures and expression variability.

using the hourglass data

using Ectocarpus data from hourglass

```
library(tidyr); library(duckplyr)
options(duckdb.materialize_message = FALSE)
```

```
# Ec_TPM <- readr::read_tsv("../atlas_ref/data/expression_data_processed/Ec_TPM.tsv") |>
#   tibble::column_to_rownames("GeneID")
Ec_TPM <- readr::read_csv("data/hourglass_data/Ec_abundance.csv") |>
  # dplyr::mutate(GeneID = stringr::str_remove(gene_id, "\\..$")) |>
  tibble::column_to_rownames("GeneID")
```

```

# check Kryuchkova-Mostacci & Robinson-Rechavi for more functions to add.
specificity <- function(abundance, measure = "tau"){
  abundance_longer <-
    abundance |>
    duckplyr::as_duckdb_tibble() |>
    dplyr::select(1, where(is.numeric)) |>
    tidyr::pivot_longer(!1, names_to = "sample", values_to = "TPM")
  if(measure == "gini"){
    breadth_out <- abundance_longer |>
      dplyr::summarise(
        gini = DescTools::Gini(TPM),
        .by = GeneID
      )
  } else if(measure == "tau"){
    n_col <- ncol(abundance) - 1
    breadth_out <- abundance_longer |>
      dplyr::summarise(
        tau = sum(1 - (TPM/max(TPM)))/(n_col-1),
        .by = GeneID
      )
  } else if(measure == "H"){
    breadth_out <- abundance_longer |>
      dplyr::summarise(
        H = DescTools::Entropy(TPM/sum(TPM)),
        .by = GeneID
      )
  } else if(measure == "rH"){
    # KL divergence from uniform distribution
    n_col <- ncol(abundance) - 1
    x_1 <-
    breadth_out <- abundance_longer |>
      dplyr::mutate(
        # rH = entropy::KL.empirical(y1 = TPM/sum(TPM), y2 = 1/n_col),
        y1 = TPM/sum(TPM),
        y2 = 1/n_col,
        .by = GeneID
      ) |>
      dplyr::summarise(
        rH = entropy::KL.empirical(y1, y2),
        .by = GeneID
      )
  } else if(measure == "mean"){
    # not really a specificity but useful nonetheless
    breadth_out <- abundance_longer |>
      dplyr::summarise(
        meanTPM = mean(TPM),
        .by = GeneID
      )
  } else {
    stop("Specify measure: \"gini\", \"tau\", \"H\", \"rH\", \"mean\"")
  }
  return(breadth_out)
}

```

```

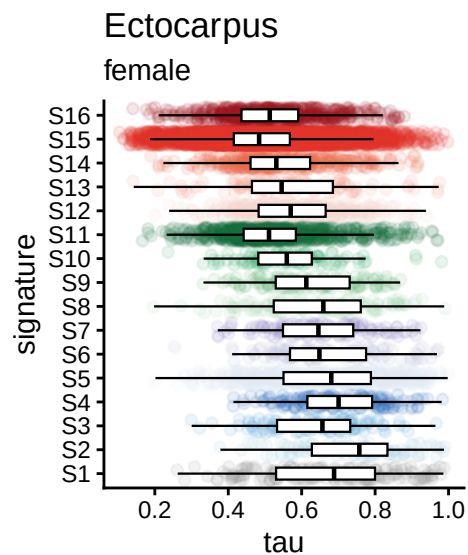
Ec_TPM.log2 <- log2(Ec_TPM + 1) |> tibble::rownames_to_column("GeneID")
# Ec_TPM.log2 <- log2(Ec_TPM[,c(7:10)] + 1) |> tibble::rownames_to_column("GeneID")

Ec_tau.log2 <- specificity(Ec_TPM.log2, measure = "tau")
Ec_H.log2 <- specificity(Ec_TPM.log2, measure = "H")
Ec_rH.log2 <- specificity(Ec_TPM.log2, measure = "rH")

Ec_combined <-
  Ec_ps.filt |>
  # dplyr::mutate(GeneID = stringr::str_remove(gene_id, "\\..$")) |>
  dplyr::rename(GeneID = gene_id) |>
  dplyr::inner_join(Ec_H.log2) |>
  dplyr::inner_join(Ec_tau.log2) |>
  dplyr::inner_join(Ec_rH.log2)

Ec_combined |>
  ggplot2::ggplot(aes(x = Signature.Female, colour = Signature.Female, y = tau)) |>
  plot_style() +
  ggplot2::labs(x = "signature", y = "tau",
                title = "Ectocarpus", subtitle = "female")

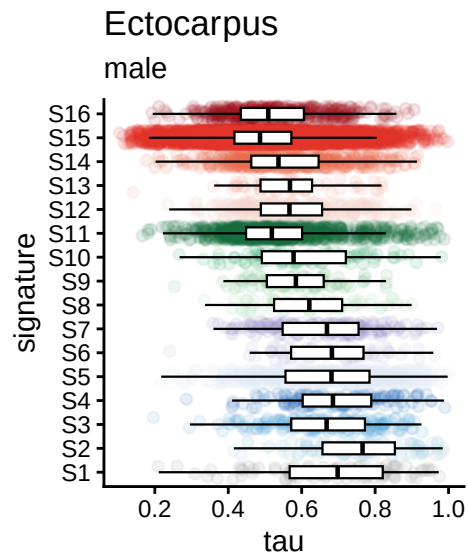
```



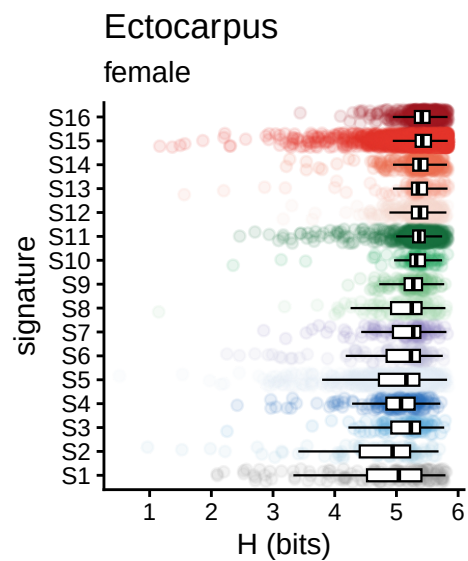
```

Ec_combined |>
  ggplot2::ggplot(aes(x = Signature.Male, colour = Signature.Male, y = tau)) |>
  plot_style() +
  ggplot2::labs(x = "signature", y = "tau",
                title = "Ectocarpus", subtitle = "male")

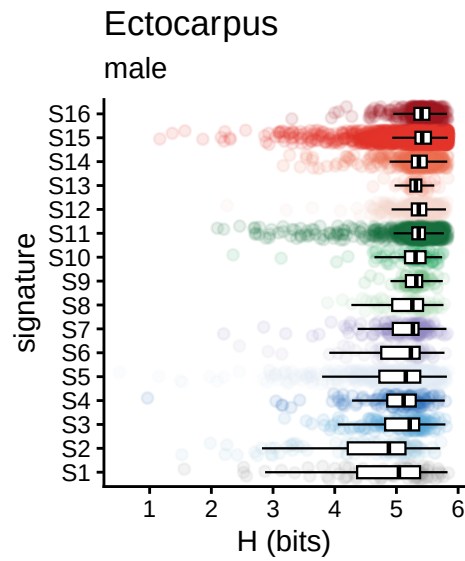
```



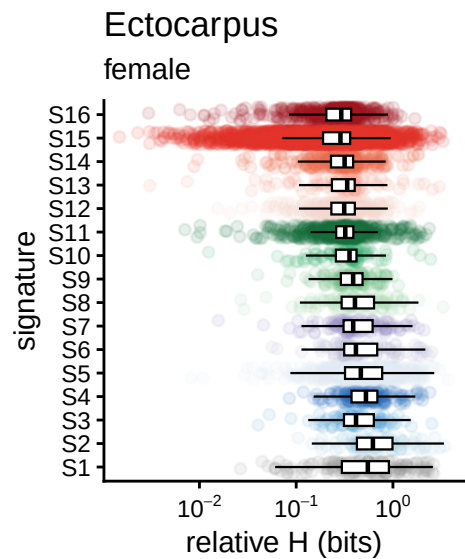
```
Ec_combined |>
  ggplot2::ggplot(aes(x = Signature.Female, colour = Signature.Female, y = H)) |>
  plot_style() +
  ggplot2::labs(x = "signature", y = "H (bits)",
    title = "Ectocarpus", subtitle = "female")
```



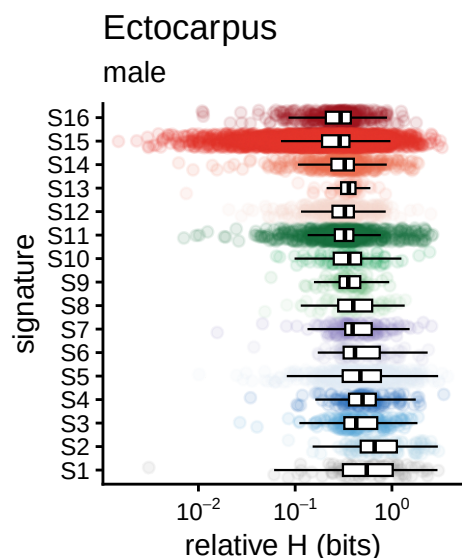
```
Ec_combined |>
  ggplot2::ggplot(aes(x = Signature.Male, colour = Signature.Male, y = H)) |>
  plot_style() +
  ggplot2::labs(x = "signature", y = "H (bits)",
    title = "Ectocarpus", subtitle = "male")
```



```
Ec_combined |>
  ggplot2::ggplot(aes(x = Signature.Female, colour = Signature.Female, y = rH)) |>
  plot_style() +
  ggplot2::scale_y_log10(labels = label_log()) +
  ggplot2::labs(x = "signature", y = "relative H (bits)",
    title = "Ectocarpus", subtitle = "female")
```



```
Ec_combined |>
  ggplot2::ggplot(aes(x = Signature.Male, colour = Signature.Male, y = rH)) |>
  plot_style() +
  ggplot2::scale_y_log10(labels = label_log()) +
  ggplot2::labs(x = "signature", y = "relative H (bits)",
    title = "Ectocarpus", subtitle = "male")
```



It should be noted in these plots that tau, H and rH were computed using all data (not just restricted to just males or females).

using Jero's RNA-seq data

```
selectGenes <- function(counts, min.count=3){
  keep <-
    counts[rowMeans(counts)>min.count, ]
  return(keep)
}
```

this is necessary because if the mean expression is 0, then we cannot compute the variability measure.
also, if mean expression is low, it might be noisy.

```
Ec_TPM.j <- readr::read_tsv("data/TPM/Ectocarpus-sp7-WT.tsv") |>
  dplyr::select(GeneID = gene_id, where(is.numeric)) |>
  tibble::column_to_rownames("GeneID") |>
  selectGenes()
Ddud_TPM.j <- readr::read_tsv("data/TPM/Desmarestia-dudresnayi.tsv") |>
  dplyr::select(GeneID = gene_id, where(is.numeric)) |>
  tibble::column_to_rownames("GeneID") |>
  selectGenes()
Dher_TPM.j <- readr::read_tsv("data/TPM/Desmarestia-herbacea.tsv") |>
  dplyr::select(GeneID = tx, where(is.numeric)) |>
  tibble::column_to_rownames("GeneID") |>
  selectGenes()
Und_TPM.j <- readr::read_tsv("data/TPM/Undaria-pinnatifida.tsv") |>
  dplyr::select(GeneID = gene_id, where(is.numeric)) |>
  tibble::column_to_rownames("GeneID") |>
  selectGenes()
Spro_TPM.j <- readr::read_tsv("data/TPM/Scytosiphon-promiscuus.tsv") |>
  dplyr::select(GeneID = tx, where(is.numeric)) |>
  tibble::column_to_rownames("GeneID") |>
  selectGenes()
Si_TPM.j <- readr::read_tsv("data/TPM/Schizocladia-ischiensis.tsv") |>
  dplyr::select(GeneID = gene_id, where(is.numeric)) |>
  dplyr::select(!dplyr::contains("RAP10_")) |>
```

```

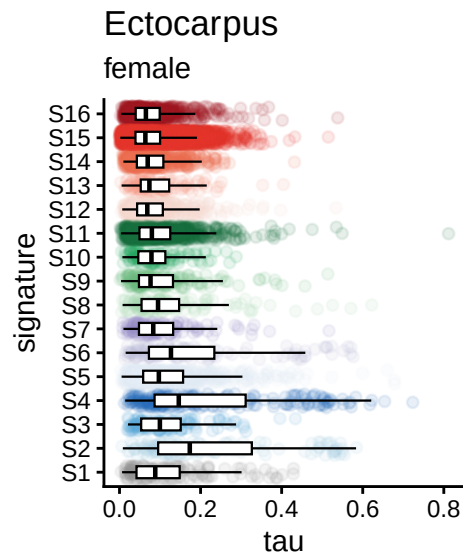
tibble::column_to_rownames("GeneID") |>
selectGenes()

Ec_TPM.j.log2 <- log2(Ec_TPM.j + 1) |> tibble::rownames_to_column("GeneID")
Ddud_TPM.j.log2 <- log2(Ddud_TPM.j + 1) |> tibble::rownames_to_column("GeneID")
Dher_TPM.j.log2 <- log2(Dher_TPM.j + 1) |> tibble::rownames_to_column("GeneID")
Und_TPM.j.log2 <- log2(Und_TPM.j + 1) |> tibble::rownames_to_column("GeneID")
Spro_TPM.j.log2 <- log2(Spro_TPM.j + 1) |> tibble::rownames_to_column("GeneID")
Si_TPM.j.log2 <- log2(Si_TPM.j + 1) |> tibble::rownames_to_column("GeneID")

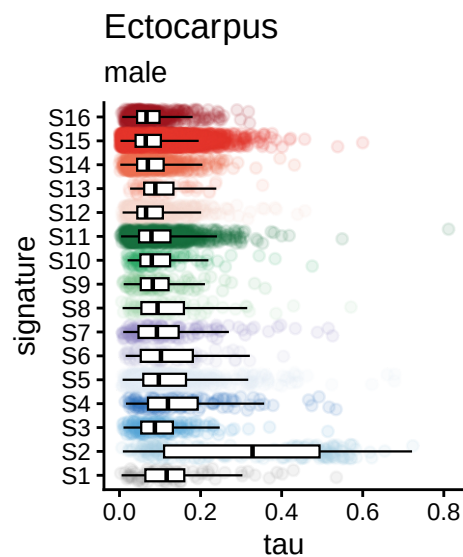
Ec_combined.j <-
  Ec_ps.filt |>
  dplyr::rename(GeneID = gene_id) |>
  dplyr::inner_join(specificity(Ec_TPM.j.log2, measure = "tau")) |>
  dplyr::inner_join(specificity(Ec_TPM.j.log2, measure = "H")) |>
  dplyr::inner_join(specificity(Ec_TPM.j.log2, measure = "rH"))
Ddud_combined.j <-
  Ddud_ps.filt |>
  dplyr::rename(GeneID = gene_id) |>
  dplyr::inner_join(specificity(Ddud_TPM.j.log2, measure = "tau")) |>
  dplyr::inner_join(specificity(Ddud_TPM.j.log2, measure = "H")) |>
  dplyr::inner_join(specificity(Ddud_TPM.j.log2, measure = "rH"))
Dher_combined.j <-
  Dher_ps.filt |>
  dplyr::rename(GeneID = gene_id) |>
  dplyr::inner_join(specificity(Dher_TPM.j.log2, measure = "tau")) |>
  dplyr::inner_join(specificity(Dher_TPM.j.log2, measure = "H")) |>
  dplyr::inner_join(specificity(Dher_TPM.j.log2, measure = "rH"))
Und_combined.j <-
  Und_ps.filt |>
  dplyr::rename(GeneID = gene_id) |>
  dplyr::inner_join(specificity(Und_TPM.j.log2, measure = "tau")) |>
  dplyr::inner_join(specificity(Und_TPM.j.log2, measure = "H")) |>
  dplyr::inner_join(specificity(Und_TPM.j.log2, measure = "rH"))
Spro_combined.j <-
  Spro_ps.filt |>
  dplyr::rename(GeneID = gene_id) |>
  dplyr::inner_join(specificity(Spro_TPM.j.log2, measure = "tau")) |>
  dplyr::inner_join(specificity(Spro_TPM.j.log2, measure = "H")) |>
  dplyr::inner_join(specificity(Spro_TPM.j.log2, measure = "rH"))
Si_combined.j <-
  Si_ps.filt |>
  dplyr::rename(GeneID = gene_id) |>
  dplyr::inner_join(specificity(Si_TPM.j.log2, measure = "tau")) |>
  dplyr::inner_join(specificity(Si_TPM.j.log2, measure = "H")) |>
  dplyr::inner_join(specificity(Si_TPM.j.log2, measure = "rH"))

Ec_combined.j |>
  ggplot2::ggplot(aes(x = Signature.Female, colour = Signature.Female, y = tau)) |>
  plot_style() +
  ggplot2::labs(x = "signature", y = "tau",
                title = "Ectocarpus", subtitle = "female")

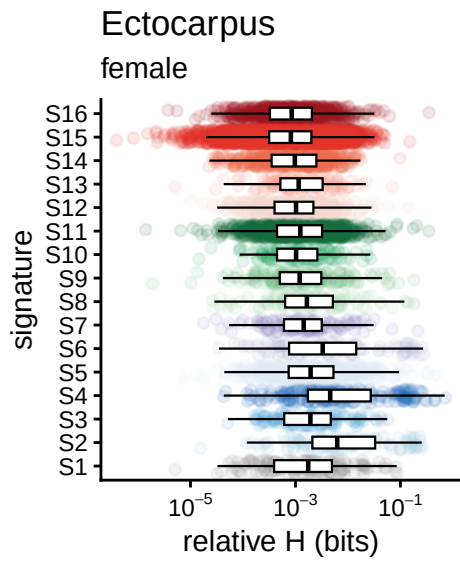
```



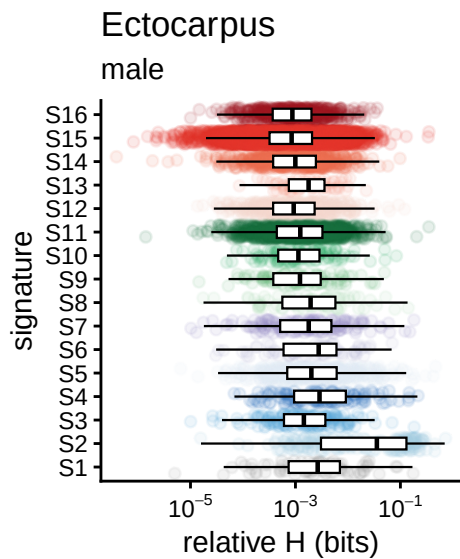
```
Ec_combined.j |>
  ggplot2::ggplot(aes(x = Signature.Male, colour = Signature.Male, y = tau)) |>
  plot_style() +
  ggplot2::labs(x = "signature", y = "tau",
                title = "Ectocarpus", subtitle = "male")
```



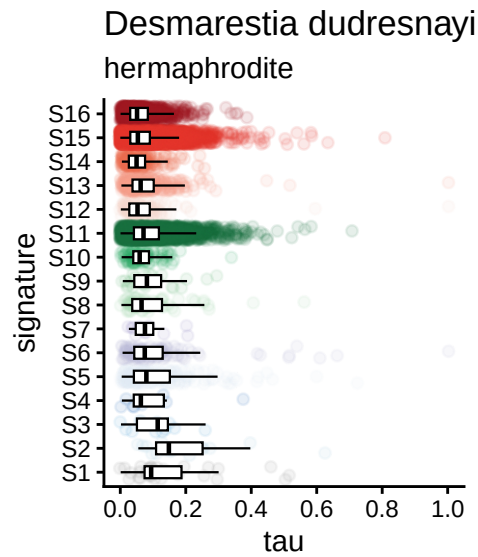
```
Ec_combined.j |>
  ggplot2::ggplot(aes(x = Signature.Female, colour = Signature.Female, y = rH)) |>
  plot_style() +
  ggplot2::scale_y_log10(labels = label_log()) +
  ggplot2::labs(x = "signature", y = "relative H (bits)",
                title = "Ectocarpus", subtitle = "female")
```



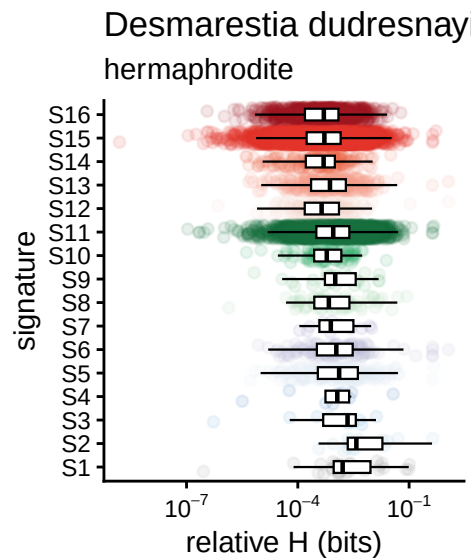
```
Ec_combined.j |>
  ggplot2::ggplot(aes(x = Signature.Male, colour = Signature.Male, y = rH)) |>
  plot_style() +
  ggplot2::scale_y_log10(labels = label_log()) +
  ggplot2::labs(x = "signature", y = "relative H (bits)",
    title = "Ectocarpus", subtitle = "male")
```



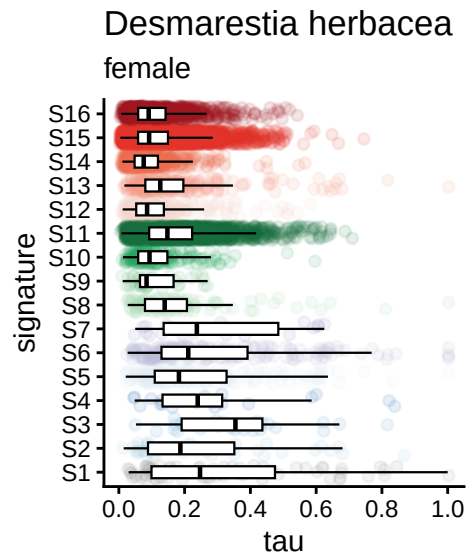
```
Ddud_combined.j |>
  ggplot2::ggplot(aes(x = Signature, colour = Signature, y = tau)) |>
  plot_style() +
  ggplot2::labs(x = "signature", y = "tau",
    title = "Desmarestia dudresnayi", subtitle = "hermaphrodite")
```



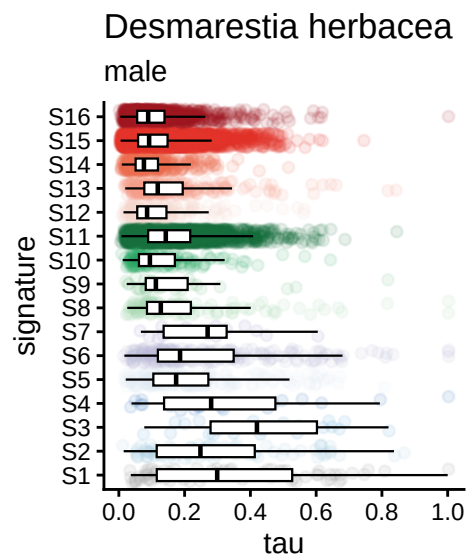
```
Ddud_combined.j |>
  ggplot2::ggplot(aes(x = Signature, colour = Signature, y = rH)) |>
  plot_style() +
  ggplot2::scale_y_log10(labels = label_log()) +
  ggplot2::labs(x = "signature", y = "relative H (bits)",
    title = "Desmarestia dudresnayi", subtitle = "hermaphrodite")
```



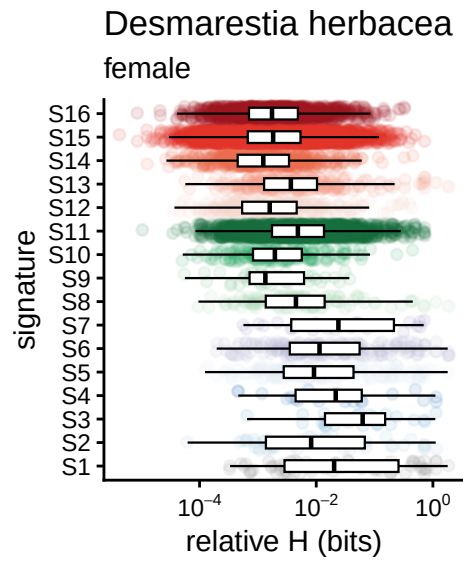
```
Dher_combined.j |>
  ggplot2::ggplot(aes(x = Signature.Female, colour = Signature.Female, y = tau)) |>
  plot_style() +
  ggplot2::labs(x = "signature", y = "tau",
    title = "Desmarestia herbacea", subtitle = "female")
```



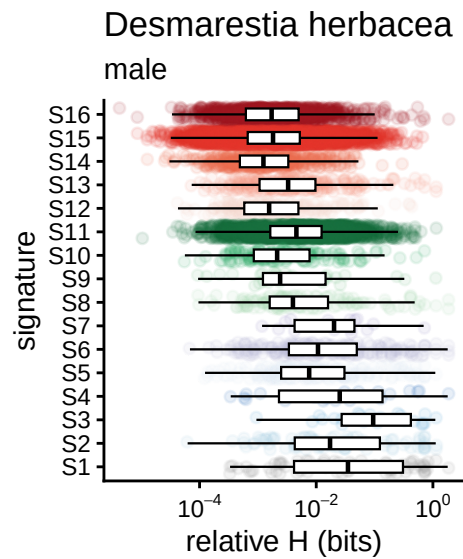
```
Dher_combined.j |>
  ggplot2::ggplot(aes(x = Signature.Male, colour = Signature.Male, y = tau)) |>
  plot_style() +
  ggplot2::labs(x = "signature", y = "tau",
    title = "Desmarestia herbacea", subtitle = "male")
```



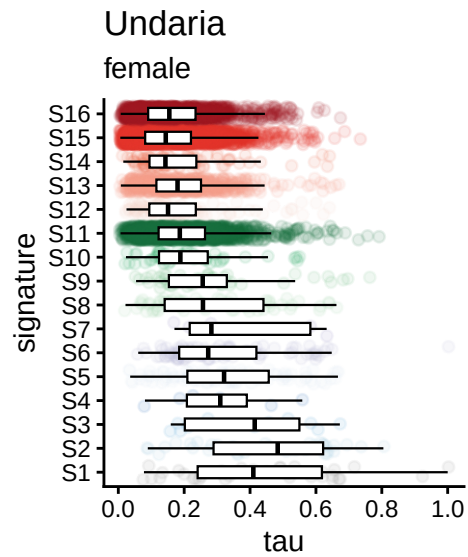
```
Dher_combined.j |>
  ggplot2::ggplot(aes(x = Signature.Female, colour = Signature.Female, y = rH)) |>
  plot_style() +
  ggplot2::scale_y_log10(labels = label_log()) +
  ggplot2::labs(x = "signature", y = "relative H (bits)",
    title = "Desmarestia herbacea", subtitle = "female")
```



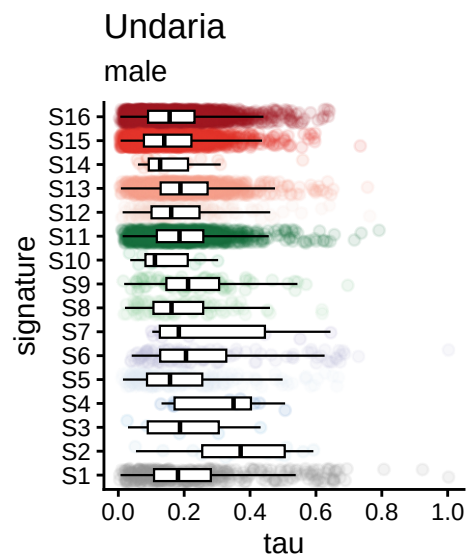
```
Dher_combined.j |>
  ggplot2::ggplot(aes(x = Signature.Male, colour = Signature.Male, y = rH)) |>
  plot_style() +
  ggplot2::scale_y_log10(labels = label_log()) +
  ggplot2::labs(x = "signature", y = "relative H (bits)",
    title = "Desmarestia herbacea", subtitle = "male")
```



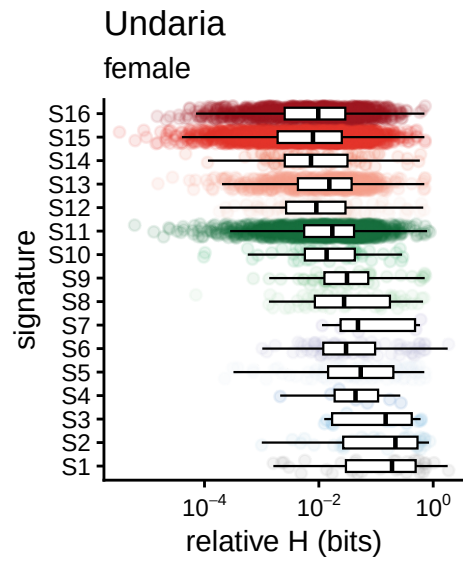
```
Und_combined.j |>
  ggplot2::ggplot(aes(x = Signature.Female, colour = Signature.Female, y = tau)) |>
  plot_style() +
  ggplot2::labs(x = "signature", y = "tau",
    title = "Undaria", subtitle = "female")
```



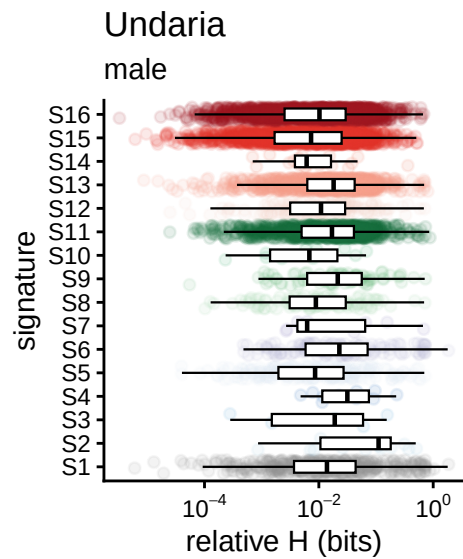
```
Und_combined.j |>
  ggplot2::ggplot(aes(x = Signature.Male, colour = Signature.Male, y = tau)) |>
  plot_style() +
  ggplot2::labs(x = "signature", y = "tau",
    title = "Undaria", subtitle = "male")
```



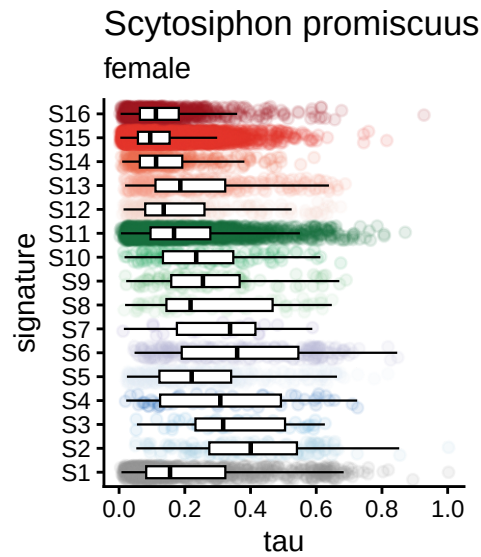
```
Und_combined.j |>
  ggplot2::ggplot(aes(x = Signature.Female, colour = Signature.Female, y = rH)) |>
  plot_style() +
  ggplot2::scale_y_log10(labels = label_log()) +
  ggplot2::labs(x = "signature", y = "relative H (bits)",
    title = "Undaria", subtitle = "female")
```



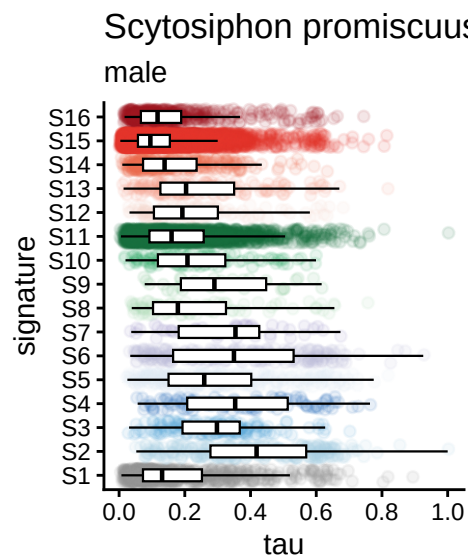
```
Und_combined.j |>
  ggplot2::ggplot(aes(x = Signature.Male, colour = Signature.Male, y = rH)) |>
  plot_style() +
  ggplot2::scale_y_log10(labels = label_log()) +
  ggplot2::labs(x = "signature", y = "relative H (bits)",
    title = "Undaria", subtitle = "male")
```



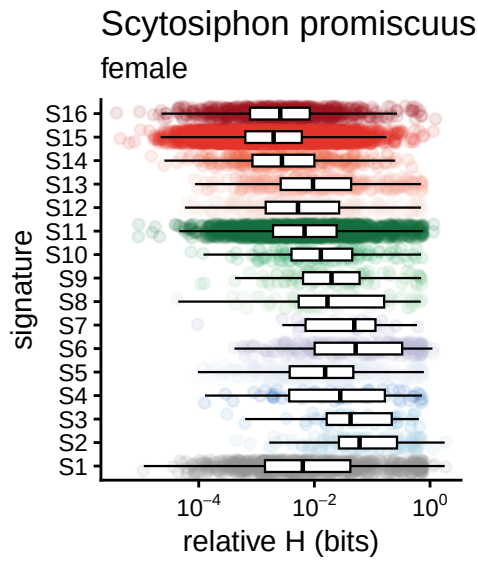
```
Spro_combined.j |>
  ggplot2::ggplot(aes(x = Signature.Female, colour = Signature.Female, y = tau)) |>
  plot_style() +
  ggplot2::labs(x = "signature", y = "tau",
    title = "Scytosiphon promiscuus", subtitle = "female")
```



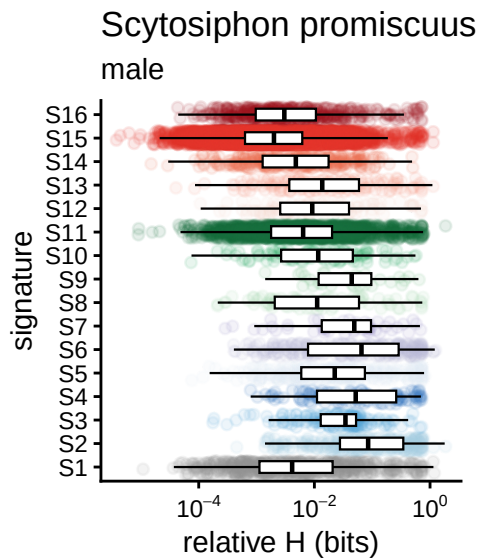
```
Spro_combined.j |>
  ggplot2::ggplot(aes(x = Signature.Male, colour = Signature.Male, y = tau)) |>
  plot_style() +
  ggplot2::labs(x = "signature", y = "tau",
    title = "Scytosiphon promiscuus", subtitle = "male")
```



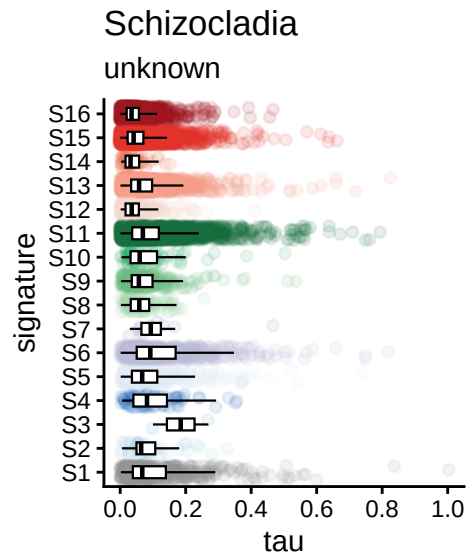
```
Spro_combined.j |>
  ggplot2::ggplot(aes(x = Signature.Female, colour = Signature.Female, y = rH)) |>
  plot_style() +
  ggplot2::scale_y_log10(labels = label_log()) +
  ggplot2::labs(x = "signature", y = "relative H (bits)",
    title = "Scytosiphon promiscuus", subtitle = "female")
```



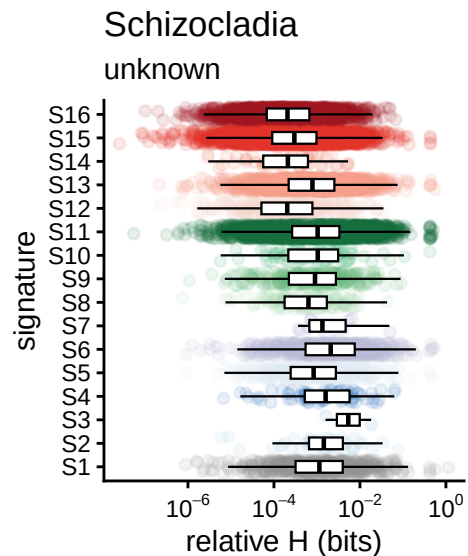
```
Spro_combined.j |>
  ggplot2::ggplot(aes(x = Signature.Male, colour = Signature.Male, y = rH)) |>
  plot_style() +
  ggplot2::scale_y_log10(labels = label_log()) +
  ggplot2::labs(x = "signature", y = "relative H (bits)",
    title = "Scytosiphon promiscuus", subtitle = "male")
```



```
Si_combined.j |>
  ggplot2::ggplot(aes(x = Signature, colour = Signature, y = tau)) |>
  plot_style() +
  ggplot2::labs(x = "signature", y = "tau",
    title = "Schizocladia", subtitle = "unknown")
```



```
Si_combined.j |>
  ggplot2::ggplot(aes(x = Signature, colour = Signature, y = rH)) |>
  plot_style() +
  ggplot2::scale_y_log10(labels = label_log()) +
  ggplot2::labs(x = "signature", y = "relative H (bits)",
    title = "Schizocladia", subtitle = "unknown")
```



It should be noted again that tau and rH were computed using all data (not just restricted to just males or females).

```
# Ec
kruskal.test(rH ~ Signature.Female, data = Ec_combined.j)

##
## Kruskal-Wallis rank sum test
##
## data:  rH by Signature.Female
## Kruskal-Wallis chi-squared = 808.23, df = 15, p-value < 2.2e-16
```

```

# Ddud
kruskal.test(rH ~ Signature, data = Ddud_combined.j)

##
## Kruskal-Wallis rank sum test
##
## data: rH by Signature
## Kruskal-Wallis chi-squared = 397.48, df = 15, p-value < 2.2e-16

# Dher
kruskal.test(rH ~ Signature.Female, data = Dher_combined.j)

##
## Kruskal-Wallis rank sum test
##
## data: rH by Signature.Female
## Kruskal-Wallis chi-squared = 1241.2, df = 15, p-value < 2.2e-16

# Und
kruskal.test(rH ~ Signature.Female, data = Und_combined.j)

##
## Kruskal-Wallis rank sum test
##
## data: rH by Signature.Female
## Kruskal-Wallis chi-squared = 483.92, df = 15, p-value < 2.2e-16

# Spro
kruskal.test(rH ~ Signature.Female, data = Spro_combined.j)

##
## Kruskal-Wallis rank sum test
##
## data: rH by Signature.Female
## Kruskal-Wallis chi-squared = 2040.6, df = 15, p-value < 2.2e-16

# Si
kruskal.test(rH ~ Signature, data = Si_combined.j)

##
## Kruskal-Wallis rank sum test
##
## data: rH by Signature
## Kruskal-Wallis chi-squared = 2135.5, df = 15, p-value < 2.2e-16

all_combined.j <-
  Dher_combined.j |>
  dplyr::select(Signature = Signature.Female, rank, rH) |>
  dplyr::mutate(sp = "Dher") |>
  rbind(Ddud_combined.j |>
    dplyr::select(Signature, rank, rH) |>
    dplyr::mutate(sp = "Ddud")
  ) |>
  rbind(Si_combined.j |>
    dplyr::select(Signature, rank, rH) |>
    dplyr::mutate(sp = "Si")
  ) |>

```

```

rbind(Spro_combined.j |>
  dplyr::select(Signature = Signature.Female, rank, rH) |>
  dplyr::mutate(sp = "Spro")
) |>
rbind(Ec_combined.j |>
  dplyr::select(Signature = Signature.Female, rank, rH) |>
  dplyr::mutate(sp = "Ec")
) |>
rbind(Und_combined.j |>
  dplyr::select(Signature = Signature.Female, rank, rH) |>
  dplyr::mutate(sp = "Und")
) |>
dplyr::summarise(
  median_rank = median(rank),
  median_rH = median(rH),
  mean_rH = mean(rH),
  .by = c(sp, Signature)
)

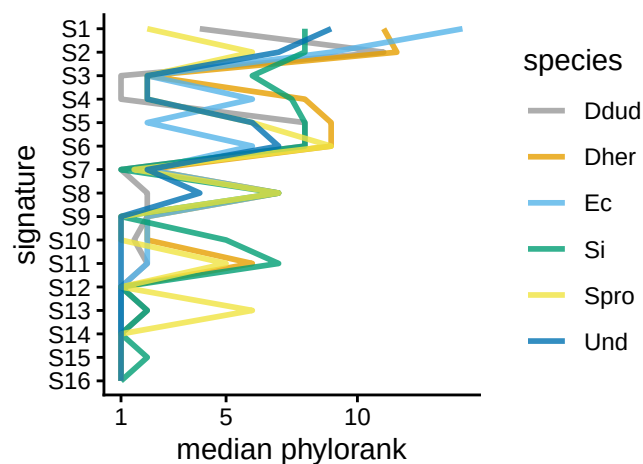
```

```
cbPalette <- c("#999999", "#E69F00", "#56B4E9", "#009E73", "#F0E442", "#0072B2", "#D55E00", "#CC79A7")
```

```

all_combined.j |>
  ggplot2::ggplot(aes(x = Signature, y = median_rank,
    colour = sp, group = sp)) +
  ggplot2::geom_line(size = 1, alpha = 0.8) +
  ggplot2::labs(x = "signature", y = "median phylorank", colour = "species") +
  ggplot2::theme_classic() +
  ggplot2::coord_flip() +
  ggplot2::scale_x_discrete(limits=rev) +
  ggplot2::scale_y_continuous(breaks = c(1,5,10)) +
  ggplot2::scale_colour_manual(values=cbPalette)

```

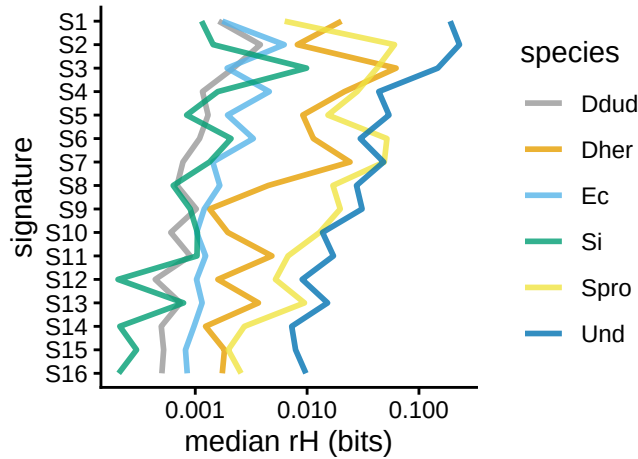


```

all_combined.j |>
  ggplot2::ggplot(aes(x = Signature, y = median_rH,
    colour = sp, group = sp)) +
  ggplot2::geom_line(size = 1, alpha = 0.8) +
  ggplot2::labs(x = "signature", y = "median rH (bits)", colour = "species") +
  ggplot2::theme_classic() +
  ggplot2::coord_flip() +

```

```
ggplot2::scale_x_discrete(limits=rev) +
ggplot2::scale_y_log10() +
# ggplot2::scale_y_continuous(breaks = scales::breaks_pretty()) +
ggplot2::scale_colour_manual(values=cbPalette)
```



session info

```
devtools::session_info()
```

```
## - Session info -----
## setting      value
## version      R version 4.4.0 (2024-04-24)
## os           macOS 15.5
## system       x86_64, darwin20
## ui           X11
## language     (EN)
## collate      en_US.UTF-8
## ctype        en_US.UTF-8
## tz           Europe/Berlin
## date         2025-10-29
## pandoc       3.4 @ /Applications/RStudio.app/Contents/Resources/app/quarto/bin/tools/x86_64/ (via rmark
##
## - Packages -----
## package      * version date (UTC) lib source
## beeswarm     0.4.0   2021-06-01 [1] CRAN (R 4.4.0)
## bit          4.6.0   2025-03-06 [1] CRAN (R 4.4.1)
## bit64        4.6.0-1 2025-01-16 [1] CRAN (R 4.4.1)
## boot         1.3-32  2025-08-29 [1] CRAN (R 4.4.1)
## cachem       1.1.0   2024-05-16 [1] CRAN (R 4.4.0)
## Cairo        1.6-5   2025-08-20 [1] CRAN (R 4.4.1)
## cellranger   1.1.0   2016-07-27 [1] CRAN (R 4.4.0)
## class        7.3-23  2025-01-01 [1] CRAN (R 4.4.1)
## cli          3.6.5   2025-04-23 [1] CRAN (R 4.4.1)
## collections  0.3.9   2025-08-18 [1] CRAN (R 4.4.1)
## crayon       1.5.3   2024-06-20 [1] CRAN (R 4.4.0)
## data.table   1.17.8  2025-07-10 [1] CRAN (R 4.4.1)
## DBI          1.2.3   2024-06-02 [1] CRAN (R 4.4.0)
```

##	DescTools	0.99.60	2025-03-28	[1]	CRAN	(R 4.4.1)
##	devtools	2.4.5	2022-10-11	[1]	CRAN	(R 4.4.0)
##	dichromat	2.0-0.1	2022-05-02	[1]	CRAN	(R 4.4.0)
##	digest	0.6.37	2024-08-19	[1]	CRAN	(R 4.4.1)
##	dplyr	* 1.1.4	2023-11-17	[1]	CRAN	(R 4.4.0)
##	duckdb	1.3.2	2025-07-09	[1]	CRAN	(R 4.4.1)
##	duckplyr	* 1.1.1	2025-07-30	[1]	CRAN	(R 4.4.1)
##	e1071	1.7-16	2024-09-16	[1]	CRAN	(R 4.4.1)
##	ellipsis	0.3.2	2021-04-29	[1]	CRAN	(R 4.4.0)
##	entropy	1.3.2	2025-04-07	[1]	CRAN	(R 4.4.1)
##	evaluate	1.0.5	2025-08-27	[1]	CRAN	(R 4.4.1)
##	Exact	3.3	2024-07-21	[1]	CRAN	(R 4.4.0)
##	expm	1.0-0	2024-08-19	[1]	CRAN	(R 4.4.1)
##	farver	2.1.2	2024-05-13	[1]	CRAN	(R 4.4.0)
##	fastmap	1.2.0	2024-05-15	[1]	CRAN	(R 4.4.0)
##	forcats	* 1.0.0	2023-01-29	[1]	CRAN	(R 4.4.0)
##	fs	1.6.6	2025-04-12	[1]	CRAN	(R 4.4.1)
##	generics	0.1.4	2025-05-09	[1]	CRAN	(R 4.4.1)
##	ggbeeswarm	0.7.2	2023-04-29	[1]	CRAN	(R 4.4.0)
##	ggplot2	* 4.0.0	2025-09-11	[1]	CRAN	(R 4.4.1)
##	ggrastr	* 1.0.2	2023-06-01	[1]	CRAN	(R 4.4.0)
##	gld	2.6.7	2025-01-17	[1]	CRAN	(R 4.4.1)
##	glue	1.8.0	2024-09-30	[1]	CRAN	(R 4.4.1)
##	gtable	0.3.6	2024-10-25	[1]	CRAN	(R 4.4.1)
##	haven	2.5.5	2025-05-30	[1]	CRAN	(R 4.4.1)
##	hms	1.1.3	2023-03-21	[1]	CRAN	(R 4.4.0)
##	htmltools	0.5.8.1	2024-04-04	[1]	CRAN	(R 4.4.0)
##	htmlwidgets	1.6.4	2023-12-06	[1]	CRAN	(R 4.4.0)
##	httpuv	1.6.16	2025-04-16	[1]	CRAN	(R 4.4.1)
##	httr	1.4.7	2023-08-15	[1]	CRAN	(R 4.4.0)
##	jsonlite	2.0.0	2025-03-27	[1]	CRAN	(R 4.4.1)
##	knitr	1.50	2025-03-16	[1]	CRAN	(R 4.4.1)
##	later	1.4.4	2025-08-27	[1]	CRAN	(R 4.4.1)
##	lattice	0.22-7	2025-04-02	[1]	CRAN	(R 4.4.1)
##	lifecycle	1.0.4	2023-11-07	[1]	CRAN	(R 4.4.0)
##	lmom	3.2	2024-09-30	[1]	CRAN	(R 4.4.1)
##	lubridate	* 1.9.4	2024-12-08	[1]	CRAN	(R 4.4.1)
##	magrittr	2.0.4	2025-09-12	[1]	CRAN	(R 4.4.1)
##	MASS	7.3-65	2025-02-28	[1]	CRAN	(R 4.4.1)
##	Matrix	1.7-4	2025-08-28	[1]	CRAN	(R 4.4.1)
##	memoise	2.0.1	2021-11-26	[1]	CRAN	(R 4.4.0)
##	mime	0.13	2025-03-17	[1]	CRAN	(R 4.4.1)
##	miniUI	0.1.2	2025-04-17	[1]	CRAN	(R 4.4.1)
##	mvtnorm	1.3-3	2025-01-10	[1]	CRAN	(R 4.4.1)
##	pillar	1.11.1	2025-09-17	[1]	CRAN	(R 4.4.1)
##	pkgbuild	1.4.4	2024-03-17	[1]	CRAN	(R 4.4.0)
##	pkgconfig	2.0.3	2019-09-22	[1]	CRAN	(R 4.4.0)
##	pkgload	1.4.0	2024-06-28	[1]	CRAN	(R 4.4.0)
##	profvis	0.4.0	2024-09-20	[1]	CRAN	(R 4.4.1)
##	promises	1.3.3	2025-05-29	[1]	CRAN	(R 4.4.1)
##	proxy	0.4-27	2022-06-09	[1]	CRAN	(R 4.4.0)
##	purrr	* 1.1.0	2025-07-10	[1]	CRAN	(R 4.4.1)
##	R6	2.6.1	2025-02-15	[1]	CRAN	(R 4.4.1)
##	RColorBrewer	1.1-3	2022-04-03	[1]	CRAN	(R 4.4.0)

```

## Rcpp                1.1.0    2025-07-02 [1] CRAN (R 4.4.1)
## readr                * 2.1.5    2024-01-10 [1] CRAN (R 4.4.0)
## readxl              * 1.4.3    2023-07-06 [1] CRAN (R 4.4.0)
## remotes             2.5.0    2024-03-17 [1] CRAN (R 4.4.0)
## rlang               1.1.6    2025-04-11 [1] CRAN (R 4.4.1)
## rmarkdown          2.29     2024-11-04 [1] CRAN (R 4.4.1)
## rootSolve          1.8.2.4  2023-09-21 [1] CRAN (R 4.4.0)
## rstudioapi         0.17.1  2024-10-22 [1] CRAN (R 4.4.1)
## S7                  0.2.0    2024-11-07 [1] CRAN (R 4.4.1)
## scales              * 1.4.0    2025-04-24 [1] CRAN (R 4.4.1)
## sessioninfo        1.2.2    2021-12-06 [1] CRAN (R 4.4.0)
## shiny              1.9.1    2024-08-01 [1] CRAN (R 4.4.0)
## stringi            1.8.7    2025-03-27 [1] CRAN (R 4.4.1)
## stringr            * 1.5.2    2025-09-08 [1] CRAN (R 4.4.1)
## tibble              * 3.3.0    2025-06-08 [1] CRAN (R 4.4.1)
## tidyr              * 1.3.1    2024-01-24 [1] CRAN (R 4.4.0)
## tidyselect         1.2.1    2024-03-11 [1] CRAN (R 4.4.0)
## tidyverse          * 2.0.0    2023-02-22 [1] CRAN (R 4.4.0)
## timechange         0.3.0    2024-01-18 [1] CRAN (R 4.4.0)
## tinytex            0.53     2024-09-15 [1] CRAN (R 4.4.1)
## tzdb                0.5.0    2025-03-15 [1] CRAN (R 4.4.1)
## urlchecker          1.0.1    2021-11-30 [1] CRAN (R 4.4.0)
## usethis             3.1.0    2024-11-26 [1] CRAN (R 4.4.1)
## vctrs               0.6.5    2023-12-01 [1] CRAN (R 4.4.0)
## vipor               0.4.7    2023-12-18 [1] CRAN (R 4.4.0)
## vroom               1.6.6    2025-09-19 [1] CRAN (R 4.4.1)
## withr               3.0.2    2024-10-28 [1] CRAN (R 4.4.1)
## xfun                0.52     2025-04-02 [1] CRAN (R 4.4.1)
## xtable              1.8-4    2019-04-21 [1] CRAN (R 4.4.0)
## yaml                2.3.10   2024-07-26 [1] CRAN (R 4.4.0)
##
## [1] /Library/Frameworks/R.framework/Versions/4.4-x86_64/Resources/library
##
## -----

```